

Facundo Ayosa

Software Developer

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Summary

Software Developer with a focus on high-performance systems and secure desktop architectures. Proven track record in engineering scalable Go backends, cross-platform desktop applications (Wails), and automated data pipelines. Expert in virtualization and container security, dedicated to building modular, local-first solutions that solve complex workflow challenges.

Experience

Junior Achievement Americas

Remote

Digital Success Manager

February 2025 - November 2025

- **Scalable AI Infrastructure:** Spearheaded the regional launch of an AI education initiative, architecting cross-functional communication protocols between IBM stakeholders and multi-territory project leads to ensure consistent program delivery.
- **Process Engineering:** Developed and deployed custom **Python automation scripts** that optimized platform workflows, resulting in a measurable increase in course completion rates with zero operational overhead.
- **Tier 3 System Reliability:** Managed distributed environment stability by resolving critical infrastructure bottlenecks, maintaining high availability for global project teams through proactive troubleshooting and 24/7 technical oversight.
- **Data Intelligence Pipelines:** Engineered high-performance **Python data analysis pipelines** to transform raw student metrics into actionable participation reports, directly informing executive stakeholder decision-making.

<https://jaamericas.org/>

355Code

Remote

Software Developer & Part-time Teacher

November 2023 - February 2025

- **Backend & API Architecture:** Designed and maintained robust backend services for complex student record systems, ensuring high availability and secure data persistence through optimized server-side logic.
- **Security & Access Control:** Orchestrated a centralized student database implementing granular **Role-Based Access Control (RBAC)**, optimizing query performance while enforcing strict data privacy standards.
- **Technical Leadership:** Collaborated with Ivy League alumni to deliver advanced curricula in **Python, Java, and JavaScript**, translating complex Computer Science fundamentals into high-impact learning outcomes for top-tier students.

<https://www.355code.com/>

Skills

Programming Languages

Python 3, JavaScript, TypeScript, Go, Java, C, Rust

Cloud & Databases

Azure, MongoDB, PostgreSQL, NoSQL, Minio, Redis

Web Technologies

React, Node.js, deno, Redux, Vite, Tailwind, TanStack, Sonner

Tools & DevOps

Debian, Alpine, Docker, Git/GitHub, Jenkins, CI/CD, Shell Scripting, Virtualization, Sandboxing/Containerization

Languages

Spanish

Native

● ● ● ● ●

English

● ● ● ● ●

Education	
Siglo 21	Graduate Certificate
Degree on Computer Science	March 2024 - Present
https://21.edu.ar/	
Junior Achievement	Microsoft Certificate
Azure Fundamentals	April 2023 - September 2023
Azure Fundamentals and Cloud Computing	
https://www.credly.com/badges/02af65bc-0ef0-4ef4-8521-8a5163b6125f/linked_in_profile	
Certifications	
Typescript: Tu completa guía y manual de mano.	May 2023
Udemy	
Teacher: Fernando Herrera.	
Full Typescript guide, with advanced type definitions.	
https://www.udemy.com/certificate/UC-a179c7a8-5a57-4b8b-a1ca-7d1a5703d507/	
JavaScript Algorithms and Data Structures	April, 2022
FreeCodeCamp	
https://www.freecodecamp.org/certification/AyosaFacundo/javascript-algorithms-and-data-structures	
Responsive Web Design	July, 2022
FreeCodeCamp	
https://www.freecodecamp.org/certification/AyosaFacundo/responsive-web-design	
DevOps with Docker: Docker basics	December 22, 2025
University of Helsinki	
Hands-on docker course with training on the basics of how to make a dockerfile.	
https://courses.mooc.fi/certificates/validate/gt2rtb3ura7n739	
DevOps with Docker: Docker Compose	December 22, 2025
University of Helsinki	
Hands-on docker compose course with training on docker-compose using multiple dockerfiles, with docker hub experience.	
https://courses.mooc.fi/certificates/validate/jn8gvqvbx5niyr	
DevOps with Docker: Security and Optimization	December 26, 2025
University of Helsinki	
Hands-on docker compose and dockerfile optimization and security course, focusing on minifying image size and improving security	
https://courses.mooc.fi/certificates/validate/y8it4a2fidirwap	
Interests	
Homelabbing	
Docker, Linux, Self-hosted, Networking, Bash scripting, Containerization, Proxmox Hypervisor, Bare Metal, ZFS, Virtualization	
Mentoring / Tutoring	
Simplifying Complex Concepts, Knowledge Sharing, Code Review, Performance Improvement, Professional Development	

Projects

Talos

April 2026 - Present

- **Cross-Platform Desktop Architecture:** Engineered a secure desktop workspace using **Go and Wails**, enabling a high-performance bridge between a **React.js** frontend and native OS capabilities for Linux and Windows.
- **Sandboxed Micro-Frontend Environment:** Developed a "Tiny App" ecosystem leveraging **scoped virtual machines** and **iframe isolation**, enforcing strict local-first security by defaulting to zero internet access and restricted directory permissions.
- **Unified SDK & Data Persistence:** Architected a **shared MySQL schema** strategy that provides isolated data persistence for internal tools while allowing modular apps to communicate through a unified **Wails SDK**.
- **Developer-Centric Observability:** Integrated a **real-time debugging console** for live log aggregation and service monitoring, facilitating community-driven development and rapid troubleshooting of internal micro-utilities.
- **Workflow Optimization:** Designed the platform to reduce context-switching by centralizing disparate team tools into a single, modular "Launchpad" focused on privacy and extensibility.

<https://github.com/ayosafacundo/Talos>

AstroForge

Dec 2025 - Present

- **Modular API Design:** Architected a high-concurrency REST API in Go utilizing a "Package by Feature" structure to eliminate circular dependencies and ensure long-term codebase maintainability.
- **Hybrid Storage Strategy:** Implemented a scalable storage architecture using MinIO for 3D Model object data with a custom versioning system and relational/NoSQL databases for metadata management, optimized for high-throughput 3D model processing.
- **Security & Middleware:** Engineered a custom middleware pipeline for centralized logging and multi-layer authentication, leveraging sub-routers to enforce strict security boundaries.

Personal Homelab

December 2024 - Present

- **Virtualization & Orchestration:** Deployed a Proxmox-based hypervisor environment to manage a fleet of LXC containers and Virtual Machines, focusing on high availability and resource optimization.
- **Network Engineering:** Established a secure, segmented network via ZeroTier VLAN and Nginx reverse proxies, managing SSL termination and firewall policies for public-facing and internal services.
- **Operational Automation:** Integrated N8N workflows and Bash-based systemd units to automate service lifecycles, monitoring, and administrative tasks across the entire infrastructure.

Pokedex

June 2022 - February 2023

- Frontend made in React using Typescript and Bun as JavaScript runtime.
- Data gathered from PokeApi V2.
- Dockerized to a 21,86MB container, and self-hosted on a Portainer LXC on my personal homelab.

<https://github.com/ayosafacundo/management-app?tab=readme-ov-file>